

Question 1 Which of the following is not in our integer sandbox? In other words, what can't we assume in COMP61?

Multiple Choice:

- (a) Integers
- (b) Addition
- (c) Subtraction
- (d) Multiplication
- (e) Division ✓
- (f) Inequalities

Question 2 Fill in the missing part of the definition from the options given:
Definition: an integer is called **even** provided it _____.

Multiple Choice:

- (a) ends with 2, 4, 6 or 8.
- (b) is not above 100
- (c) can be written as $2x$ for some number x
- (d) is divisible by 2 ✓

Question 3 Fill in the missing parts of the definition from the options given:
Divisible: a is divisible by b if there exist a /an _____ such that _____.

Multiple Choice:

- (a) number c ; $c = b$
- (b) number c ; b times c does not equal a
- (c) integer c ; b times c equals a ✓

- (d) number c ; b times c equals a
-

Question 4 Fill in the missing parts of the definition from the options given:
Odd: An integer a is called odd provided there is _____ such that _____.

Multiple Choice:

- (a) a rational number x ; $a = 2x$
 - (b) an integer x ; $a = 2x + 1$ ✓
 - (c) an integer x ; $a = x$
 - (d) any number x ; $a = 2x + 1$
-

Question 5 Fill in the missing parts of the definition from the options given:
Prime: x is prime provided that _____ and the _____ of x is/are _____.

Multiple Choice:

- (a) $x > 0$; only positive divisor; 1
 - (b) $x > 0$; only divisor; 1
 - (c) $x > 1$; only divisors; 1 and x
 - (d) $x > 1$; only positive divisors; 1 and x ✓
-

Question 6 What are the correct ways to say a is divisible by b ?

Select All Correct Answers:

- (a) b divides a ✓
 - (b) b is a factor of a ✓
 - (c) a is a divisor of b
 - (d) b is a divisor of a ✓
 - (e) $b \mid a$ ✓
-